

Qupath Workshop

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Motivations for holding this workshop

- I agreed to host an imaging tutorial, and as I approach my 2 remaining months of PhD time, Qupath stood out as the easiest analysis tool for me to explain
- Using Qupath been very well suited for my own analysis needs, and helped me get more data out of my own images, I hope you can all pick up some ideas for your own projects 😊

Background

- Qupath is open source and runs on windows and mac
- Handles large slide scanner images well (pyramidal file formats), so great for the .vsi files from the VS200
- No need to convert to .tif first (saves time), but you can open those too if you want
- Organises images in projects, good for keeping track of a large imaging project!
- I find the 'logic' of this software easily makes sense to me
 - First you make 'objects' e.g. tissue areas, cells
 - Then you measure them (e.g. channel intensity of each cell, cell type classification)
 - Then you look at the measurements (export for fancier analysis in R)
- Large community of users, so plenty of help online available

Resources

- [Qupath installation](#)
- [Cellpose for Qupath extension](#)
- <https://forum.image.sc/tag/qupath> : Image.sc forum has a really great community of Qupath users. If you have an issue, chances are someone already has a solution for you on the forum
- [Qupath for fiji fans on youtube](#)
- [From Samples to Knowledge 2025](#) This is a really comprehensive workshop for Qupath, check all the helpful links there
- [Also the one from 2023 on youtube](#)

Content overview

- I will not be showing the install, instructions on the website
- I am using a slightly older version of Qupath, but what I'm presenting should hold for newer versions (if anything even easier with newer versions)
- Opening VS200 .vsi images directly in Qupath
- Outlining tissue areas of interest
- Segmenting cells with cell pose
- Train a classifier to classify cell types (random trees)
- Sending areas to ImageJ to make nice images for lab book/presentation
- Measuring cell distances from tissue areas
- Exporting cell measurements
- Bonus treat: using the new version of Qupath's segmentation (it works really well!)